Isaac Mubiru

isaacmubiru99@gmail.com | linkedin.com/in/isaac-mubiru | github.com/zac-09 |

Kampala, Uganda · EAT (UTC+3, 2h ahead of Geneva) · Remote

PROFESSIONAL SUMMARY

Backend software engineer with **6+ years building backend services and APIs in Node.js**, including production **TypeScript** APIs and Python data-migration tooling. Architected an **event-driven microservices backend on Apache Kafka, Docker and Kubernetes** for an NGO in Uganda, built a real-time two-way sync service on Google Pub/Sub, and led a **zero-downtime migration of 20+ production applications** to GCP reporting to the CTO. Built **Web3/blockchain backends** for DeFi applications and a USSD service that reached users on basic phones. Five years working remotely and independently for US and Singapore teams, now applying the same backend discipline to verified connectivity for schools in East and Southern Africa. Code: github.com/zac-09

CORE COMPETENCIES

Backend Services & REST APIs (Node.js, TypeScript) | Event-Driven Systems & Apache Kafka | Distributed & Service-Oriented Applications | Asynchronous Processing & Message Queues | JSON-Based Third-Party Integrations | Containerized Deployment (Docker, Kubernetes) | Blockchain / Web3 Backends | Python Data & Migration Scripts

WORK EXPERIENCE

MTailor

Sep 2022 – Present

Software Engineer (US Remote, reports to CTO)

- Implemented **real-time two-way sync between MongoDB and Firestore** using Node.js and **Google Pub/Sub** for asynchronous message processing, keeping both datastores serving live production traffic during the migration.
- Led the end-to-end migration of **20+ applications** from Parse/MongoDB to Firebase/GCP with **zero downtime**.
- Saved the company **$5,000/month** by migrating all services off AWS.
- Migrated file storage from Amazon S3 to Google Cloud Storage via a **Python** script; migrated a Webflow website from a Parse backend to Firebase.
- Implemented Express Shipping, generating an additional **$40 revenue per order**; built a 3D visualisation feature using video overlay and ffmpeg, increasing buyer conversion.

Stack: Node.js, JavaScript, Python, Firebase/Firestore, MongoDB, GCP Pub/Sub, AWS S3/EC2/EBS, ffmpeg, React

MTailor (Contractor)

Mar 2022 – Jul 2022

Backend Migration Contractor (US Remote)

- Built a migration script reading data from MongoDB and writing to Firestore after running complex processing logic.
- Wrote documentation for fellow engineers on the new Firebase SDKs (Node.js and Python); trained the Ops team on the Firebase dashboard and backup procedures.

Stack: Node.js, JavaScript, Python, MongoDB, Firebase, GCP, AWS

Mind2matter

Jan 2022 – Jul 2022

Software Engineer (US Remote, reports to CTO)

- Built **backends for DeFi applications using Web3 and Node.js** for a US agency with demanding clients.
- Delivered React UIs from Figma designs under tight timelines.

Stack: Node.js, JavaScript, Web3, Firebase, React

Dr Wealth

Aug 2021 – Jan 2022

Software Engineer (Singapore Remote)

- Extended **backend APIs in TypeScript** using Firebase Cloud Functions and Express.js hosted on Heroku.
- Built jobs querying over **2 million Firestore records** to keep customer prices current from the **Morningstar APIs** on a real-time PWA.
- Added responsive UI pages to the PWA using React, Tailwind and HTML.

Stack: Node.js, TypeScript, Express.js, Firebase, React, Tailwind

CodeBits

Jan 2020 – Jul 2021

Software Engineer, Team Lead (Uganda Remote)

- Architected an **event-driven microservices backend using Apache Kafka, Docker and Kubernetes** for FIDA Uganda's case management app.
- Built a **USSD service** for legal-aid providers communicating with a Node.js backend via **gRPC**, reaching users without smartphones.
- Led a team of 4 developers building legal-tech systems for NGOs: a case management system for FIDA Uganda and a paralegal database for LASPNET Uganda.
- Developed a cross-platform mobile app for LASPNET with React Native and Expo (iOS and Android) and Firebase push notifications.

Stack: Node.js, Apache Kafka, NATS Streaming, gRPC, Docker, Kubernetes, React Native, Firebase, USSD

PROJECTS

Event-Driven Microservices Backend (FIDA Uganda) Kafka + Kubernetes

Case-management backend for FIDA Uganda, split into services around Kafka events, containerized with Docker, deployed on Kubernetes, with gRPC for synchronous calls. Served web, mobile and USSD clients from one set of services.

Node.js, Apache Kafka, NATS Streaming, gRPC, Docker, Kubernetes

Real-Time MongoDB ↔ Firestore Sync (MTailor) Asynchronous processing

Node.js sync service on Google Pub/Sub that kept two production datastores consistent in both directions while 20+ applications were cut over one by one with zero downtime.

Node.js, Google Pub/Sub, MongoDB, Firestore, GCP

USSD Legal-Aid Service (LASPNET / CodeBits) Low-connectivity users

USSD front end for legal-aid providers backed by a Node.js service over gRPC, built for users on basic phones with no data connection.

Node.js, gRPC, USSD

EDUCATION

B.Sc. Computer Engineering — Makerere University Aug 2018 – Sep 2022

Kampala, Uganda

SKILLS

Proficient: Node.js, TypeScript, JavaScript, Python, Firebase, GCP, MongoDB, SQL, Git, Linux, React, HTML, CSS
Backend & infrastructure: Apache Kafka, NATS Streaming, Google Pub/Sub, gRPC, Express.js, Docker, Kubernetes, AWS S3/EC2/EBS
Intermediate: Go, AWS, Rust, React Native, Web3
Languages: English (fluent), Kiswahili